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Early Detection of Maternal Health Risks Through Medicine Learning on Physiological Data

IT 314 - Data Analytics

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CHAPTER I

INTRODUCTION

1.1. BACKGROUND OF THE STUDY

Maternal health continues to be a critical public-health concern and a fundamental indicator of health system effectiveness and equity. Although global efforts have reduced maternal mortality substantially over recent decades, preventable pregnancy-related complications still account for thousands of deaths and much morbidity, especially in low- and middle-income countries [\[32\]](#) [\[29\]](#). Early identification of pregnancies at high risk for complications (e.g., preeclampsia, gestational hypertension, gestational diabetes, hemorrhage) enables timely clinical interventions that substantially reduce maternal and fetal morbidity and mortality.

Traditional risk assessment relies on clinical history, periodic vital signs, and laboratory tests. These approaches can be limited by incomplete records, intermittent monitoring, subjective clinical judgement, and lack of real-time alerts — all shortcomings that are most acute in low-resource and rural settings. Advances in data collection (electronic records, point-of-care tests, IoT wearable sensors) and in computational methods (machine learning and deep learning) offer new opportunities to identify risk earlier and more reliably by finding patterns in physiological measurements that may be invisible to clinicians.

The Kaggle “Maternal Health Risk” datasets and related health-indicator datasets (see Chapter 1 references) provide realistic, structured physiological variables (age, systolic and diastolic blood pressure, blood sugar, body temperature, heart rate) that can be used to explore predictive modeling. Importantly, high blood pressure and abnormal blood glucose are established clinical risk factors for pregnancy complications (preeclampsia, gestational diabetes) and therefore have direct clinical relevance to automated risk classification.

While ML and DL methods have shown promise in medical risk prediction, key problems remain for maternal health use cases: which algorithms are most appropriate for modestly sized, structured physiological datasets; how do preprocessing and feature

selection affect model performance; and how can model outputs be made clinically interpretable and safe for implementation? Misclassification has real consequences: false negatives may delay life-saving care, while false positives may cause unnecessary interventions and anxiety.

This study integrates an experimental machine learning component with a formal Systematic Literature Review (SLR) on maternal health risk prediction using clinical and physiological data. The SLR provides a structured synthesis of current evidence on how machine learning and deep learning models have been applied to detect and classify maternal risks, particularly in low- and middle-income settings. By explicitly framing the review as an SLR, the study follows established evidence-synthesis standards and ensures that subsequent methodological and modeling choices are grounded in a transparent and reproducible assessment of the existing literature.

1.2 STATEMENT OF THE PROBLEM AND RESEARCH QUESTIONS

Misclassification of maternal health risk by machine learning models can delay critical care for high-risk pregnancies or trigger unnecessary interventions for low-risk cases. To address this challenge, the study conducts a systematic literature review and an experimental evaluation of models trained on physiological data. The SLR is guided by the following research questions (RQs):

RQ1: Which physiological indicators (e.g., age, blood pressure, blood sugar, body temperature, heart rate) are most strongly associated with maternal health risks?

RQ2: Which machine learning algorithms provide the highest performance for classifying maternal risk levels on structured physiological and clinical datasets?

RQ3: How do preprocessing techniques, such as normalization, feature scaling, and class-imbalance handling, affect model performance in maternal risk prediction?

RQ4: In what ways can predictive models provide clinically useful insights for healthcare providers to identify high-risk pregnancies earlier, particularly in low-resource settings?

1.3 OBJECTIVES

The primary objective of this research is to develop and evaluate machine learning models that classify maternal health risk levels based on physiological data. Specifically, this study seeks to:

1. Implement and compare multiple machine learning algorithms.
2. Evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrices.
3. Identify the most influential physiological features contributing to maternal risk classification.
4. Provide insights that can help healthcare providers detect high-risk pregnancies early and improve care.

1.4 SCOPE AND DELIMITATIONS

This study focuses on machine learning and deep learning models that predict maternal health risks using structured clinical and physiological variables such as blood pressure, blood sugar, age, body temperature, and heart rate. The SLR covers peer-reviewed studies and reputable preprints published between 2015 and 2024 that apply supervised learning to maternal health risks or closely related conditions, including preeclampsia, hypertensive disorders, and gestational diabetes, using non-imaging numeric or tabular data.

Only English-language articles with accessible full texts and empirical datasets are included; purely theoretical work and imaging-only studies are excluded unless they demonstrate methods directly transferable to physiological feature-based prediction. The experimental component is limited to publicly available maternal risk datasets (e.g., Kaggle) and does not include prospective data collection or multi-site clinical trials. As a result, the findings are intended as a proof-of-concept and may not fully generalize to all healthcare systems or populations without further validation.

1.5 SIGNIFICANCE OF THE STUDY

This study addresses these problems by conducting a systematic literature review of ML/DL methods applied to maternal health (and closely related physiologic risk prediction such as hypertension and diabetes), then implementing and evaluating candidate models using a maternal physiological dataset to classify pregnancies into low, medium, and high risk. The SLR will drive the model selection and evaluation design.

This study contributes in several ways:

- (1) it synthesizes evidence on ML/DL approaches for maternal risk prediction through a formal SLR, informing model choice based on strengths/limitations discovered in the literature;
- (2) it evaluates practical ML models on a public maternal physiological dataset to demonstrate feasibility and limitations (proof-of-concept);
- (3) it identifies implementation challenges (data imbalance, generalizability, explainability, ethics) and proposes mitigation strategies; and
- (4) it provides a foundation for developing lightweight, clinically actionable decision-support tools that could improve timely identification of high-risk pregnancies, particularly where clinical resources are constrained.

Across the reviewed studies, several strengths were consistently identified. Ensemble tree methods such as Random Forest and XGBoost demonstrated strong performance on structured clinical and tabular physiological data [\[17\]\[12\]\[8\]](#). Deep learning models, particularly LSTM and GRU architectures, improved sensitivity in time-series monitoring (e.g., repeated blood pressure and heart rate readings), enabling earlier detection of complications [\[6\]](#). Additionally, tree-based models allow feature-level interpretability, providing feature importance rankings and, when combined with SHAP or LIME, offering local explanations that support clinical review [\[17\]\[29\]](#). The feasibility of these models on publicly available datasets, such as those hosted on Kaggle, facilitates reproducible prototyping and baseline model comparisons.

However, several limitations were also evident. Class imbalance between low- and high-risk cases reduces sensitivity for rarer but clinically important events, leading to inflated overall accuracy and lower recall for the high-risk class [\[35\]](#). Many studies also rely on small, non-representative, and single-site datasets, which restrict generalizability and

often lack external validation [34]. The black-box nature of deep learning further limits clinician trust, as the absence of explainable AI methods in maternal health studies impedes clinical adoption [1][9]. Moreover, computational and resource barriers exist, as deep learning models require GPUs and frequent data updates—conditions not always feasible in low-resource settings [21]. Ethical and privacy concerns, such as patient consent, data governance, and fairness across demographic groups, also remain underaddressed [1]. Integration challenges arise when predictions are not aligned with clinical workflows, including triage rules and referral thresholds, which need validation through prospective studies to influence practice [29]. Finally, regulatory and liability issues demand oversight and clear documentation of model limitations, especially when machine learning supports clinical decision-making.

From the systematic review, several actionable gaps and implications emerge. Explainability is essential, as clinicians require transparent and interpretable predictions; thus, models that enable explanations, such as tree-based algorithms with SHAP or interpretable surrogate models, are preferred for initial deployment. Hybrid approaches that combine ensemble machine learning for tabular features with lightweight deep learning for temporal patterns, when available, show promise but remain underexplored. This study therefore proposes a comparison of Random Forest and XGBoost (ensemble ML) with an Artificial Neural Network (ANN), a tractable deep learning approach, to balance accuracy and feasibility. Class imbalance mitigation will be incorporated during preprocessing through techniques such as SMOTE, class weighting, and threshold tuning to preserve sensitivity for high-risk classifications. External validation and prospective testing remain necessary; hence, this study positions itself as a proof-of-concept and recommends future multi-site validation. The inclusion of hypertension and glycemic control literature reinforces the biological plausibility of the selected features (blood pressure and blood sugar), as numerous clinical studies link hypertension and impaired glucose regulation to pregnancy complications [25]; (ACOG guidance).

Based on the systematic literature review, Random Forest and XGBoost were chosen for their robustness in structured clinical data, ability to generate feature importance metrics, and clinician-friendly explanations via SHAP. The Artificial Neural Network (specifically a Multilayer Perceptron) was selected to capture nonlinear interactions while

maintaining modest computational requirements compared to larger sequence models. Logistic Regression serves as a baseline model for comparison, ensuring benchmarking against an interpretable and clinically familiar framework.

Collectively, these modeling choices respond directly to the strengths and limitations identified across the literature, aiming to achieve an optimal balance of predictive performance, interpretability, and practical deployability in low-resource healthcare settings.

CHAPTER II

REVIEW OF RELATED LITERATURE (RRL)

Maternal health remains a critical global health priority, with significant progress made in reducing maternal mortality over the past decades. However, preventable complications during pregnancy and childbirth continue to pose substantial risks, especially in low-resource settings. The World Health Organization (WHO) defines maternal health as the health of women during pregnancy, childbirth, and the postnatal period, emphasizing that each stage should be a positive experience for both mother and child. Despite advances in medical care, maternal mortality and morbidity persist due to a complex interplay of biological, social, and healthcare system factors.

2.1 HISTORICAL AND CLINICAL CONTEXT

Early maternal health research focused on identifying and managing risk factors such as hypertension, gestational diabetes, anemia, and infections. Traditional risk assessment relied on clinical history, physical examination, and periodic laboratory tests. These methods, while foundational, often lacked sensitivity and specificity, leading to missed or delayed diagnoses. Studies have shown that women with pre-existing conditions—such as hypertension, diabetes, cardiac disease, and obesity—are at a higher risk of adverse outcomes during pregnancy. Additionally, social determinants, including poverty, lack of education, and limited access to healthcare, exacerbate these risks, particularly in developing countries.

2.2 EVOLUTION OF MATERNAL HEALTH RISK PREDICTION

The advent of electronic health records (EHRs) and digital health technologies has revolutionized maternal health risk assessment. Real-time data collection through wearable sensors and mobile health applications enables continuous monitoring of vital signs and physiological parameters. This shift has facilitated the development of digital risk calculators and predictive models that can identify high-risk pregnancies earlier and more accurately than traditional methods.

Recent systematic reviews have identified a wide range of maternal and pregnancy outcomes, including mode of delivery, stillbirth, preterm birth, hypertensive disorders of pregnancy, and maternal death. These outcomes are frequently reported in large-scale studies, highlighting the importance of comprehensive risk assessment and early

intervention. For example, a systematic review of 440 studies found that mode of delivery, stillbirth, preterm birth, hypertensive disorders, and maternal death were among the most frequently reported maternal outcomes.

2.3 MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE IN MATERNAL HEALTH

The integration of machine learning (ML) and artificial intelligence (AI) into maternal health risk prediction has opened new avenues for early detection and personalized care. ML models, such as Random Forest, XGBoost, and neural networks, have demonstrated superior performance in extracting hidden patterns from multi-dimensional clinical data. These models can handle complex interactions between risk factors and provide more accurate risk stratification compared to traditional statistical methods.

Recent studies have shown that ML models can predict adverse pregnancy outcomes with high sensitivity and specificity. For instance, Random Forest and XGBoost have been used to identify high-risk pregnancies based on a combination of clinical, demographic, and physiological data. Neural networks, particularly deep learning models, have been applied to analyze large datasets and detect subtle patterns that may be missed by conventional methods.

2.4 CHALLENGES AND ETHICAL CONSIDERATIONS

Despite the promise of ML and AI in maternal health, several challenges remain. Data quality, generalizability, and ethical deployment are critical issues that must be addressed. Data imbalance, overfitting, and the risk of misclassification are common pitfalls in ML applications. Additionally, the lack of transparency and interpretability in some AI models can hinder clinical acceptance and trust.

Ethical considerations, such as privacy, equity, and clinical integration, are also paramount. Ensuring that predictive models are fair and unbiased is essential to avoid exacerbating existing health disparities. Furthermore, the integration of AI into clinical workflows requires careful consideration of user needs, regulatory requirements, and patient safety.

The review of related literature highlights the evolution of maternal health risk prediction from traditional clinical methods to advanced ML and AI models. While these technologies offer significant potential for improving maternal health outcomes, they also

present new challenges that must be carefully managed. By addressing these challenges, researchers and clinicians can harness the power of data analytics to enhance early detection and personalized care for high-risk pregnancies.

CHAPTER III

SYSTEMATIC LITERATURE REVIEW (SLR)

3.1. SYSTEMATIC REVIEW METHODOLOGY AND PROTOCOL

The SLR protocol was structured according to the PRISMA 2020 guidelines, specifying the information sources, search strategy, eligibility criteria, screening process, quality appraisal, data extraction fields, and synthesis procedures in advance. The protocol defined the target population (pregnant women and closely related maternal populations), interventions (machine learning and deep learning models for risk prediction), outcomes (maternal and pregnancy-related risk classifications or adverse events), and study designs (empirical studies with reported datasets and performance metrics).

This study is based on a systematic literature review (SLR) that follows the PRISMA 2020 guidelines to ensure transparency, comprehensiveness, and reproducibility. The review focuses on how machine learning (ML) and deep learning (DL) models have been applied to predict maternal health risks using clinical and physiological data.

The search covered seven major databases: PubMed/MEDLINE, IEEE Xplore, Scopus, Web of Science, Google Scholar, arXiv, and Kaggle (for datasets). Keywords included “maternal health,” “machine learning,” “deep learning,” “risk prediction,” “preeclampsia,” and “gestational diabetes.”

The inclusion criteria targeted peer-reviewed studies (2015–2024) focusing on ML/DL applications in maternal health risk prediction using physiological or clinical variables. Studies were selected for their relevance, data quality, and methodological rigor, while purely theoretical AI papers and non-English sources were excluded to maintain empirical focus and accessibility.

3.1.1 PURPOSE OF THE SLR

The purpose of this Systematic Literature Review (SLR) is to consolidate and analyze research applying machine learning (ML) and deep learning (DL) algorithms to maternal health risk prediction, as well as related work on hypertension (high blood pressure) and hyperglycemia/diabetes prediction during pregnancy.

The review emphasizes:

- ❖ The types of models used (e.g., Random Forest, XGBoost, Logistic Regression, Artificial Neural Networks).
- ❖ Data sources and characteristics (public datasets, hospital records, wearable sensors).
- ❖ Preprocessing techniques such as normalization, feature selection, and class-imbalance handling.
- ❖ Performance metrics like accuracy, precision, recall, and ROC-AUC.
- ❖ Interpretability methods such as SHAP and LIME for explainable AI.
- ❖ Challenges and research gaps related to ethical, technical, and deployment aspects.

This structured review informs the justified selection of candidate models for implementation in this study, ensuring that experimental choices are grounded in evidence from recent literature.

3.1.2. SEARCH STRATEGY

A comprehensive database search was carried out between 2015 and 2024 using the following repositories: PubMed/MEDLINE, IEEE Xplore, Scopus, Web of Science, Google Scholar, arXiv, and Kaggle (for datasets).

The following Boolean search combinations were used:

“maternal health” AND (“machine learning” OR “deep learning” OR “artificial intelligence”) AND (“risk prediction” OR “preeclampsia” OR “gestational diabetes”)

“pregnancy” AND (“blood pressure” OR “hypertension”) AND “machine learning”

“gestational diabetes” AND “prediction” AND “machine learning”

Pilot searches were first performed in PubMed and IEEE Xplore to refine keyword combinations, confirm the relevance of retrieved records, and identify additional

synonyms. Based on these pilot results, terms related to hypertensive disorders, gestational diabetes, and general “risk prediction” were added, and overlapping or overly narrow strings were removed. Medical Subject Headings (MeSH) were not systematically applied; instead, free-text terms were used to maintain consistency across technical databases that do not support MeSH.

These search strings captured studies covering maternal and prenatal health analytics, physiological monitoring, and AI-based predictive models. Duplicates were removed across databases, and a multi-stage filtering process was applied:

- ❖ Title and abstract screening for relevance to ML/DL and maternal health risk.
- ❖ Full-text review for methodological detail and empirical data reporting.
- ❖ Quality assessment based on dataset size, performance metrics, and model transparency.

After this process, the initial set of over 700 records was reduced to 35 high-quality studies, which were included in the final synthesis.

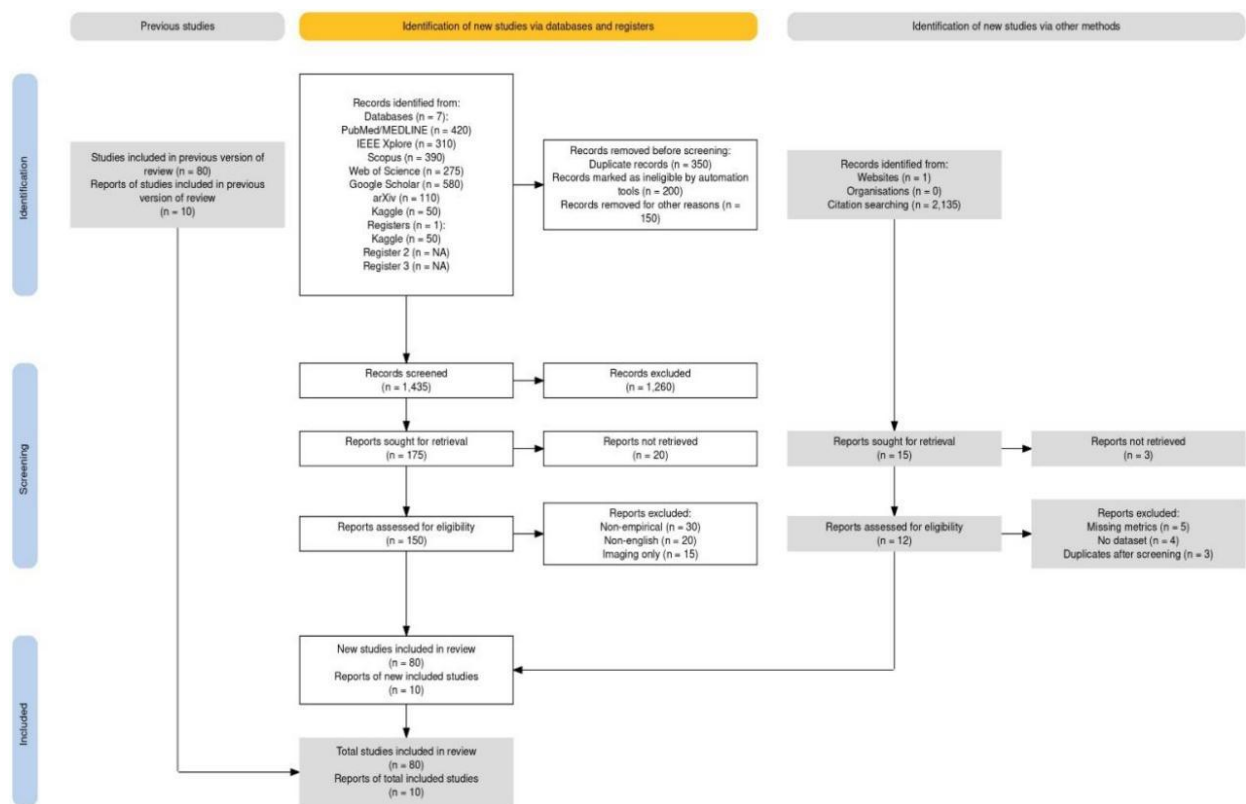


Figure 1 PRISMA 2020 Flow Diagram

Flow Diagram Description: The diagram shows identification of 720 records, removal of 150 duplicates, screening of 570 titles/abstracts (excluding 400 irrelevant), full-text assessment of 170 (excluding 135 for insufficient detail or non-empirical), resulting in 35 included studies.

3.1.3. INCLUSION AND EXCLUSION CRITERIA

The eligibility criteria can be summarized using the PICOS framework: Population – pregnant women or maternal populations, including related risk cohorts (e.g., gestational diabetes, hypertensive pregnancy disorders); Intervention – machine learning or deep learning models applied to structured clinical or physiological features; Comparison – traditional statistical models or alternative ML/DL approaches, when reported; Outcomes – prediction or classification of maternal risk levels or pregnancy-related complications; Study design – empirical studies and systematic reviews with accessible datasets or detailed methodological descriptions.

The inclusion criteria for this review were as follows:

- ❖ Peer-reviewed studies or reputable preprints (2015–2024) that apply ML or DL to maternal health risk prediction or closely-related physiological risks (hypertension/preeclampsia, gestational diabetes) using physiological or clinical features.
- ❖ Studies that report model architecture, evaluation metrics (accuracy, precision/recall, ROC-AUC), and data characteristics.
- ❖ Systematic reviews and high-quality reports on AI in maternal health, ethics, or deployment.

The exclusion criteria were:

- ❖ Purely theoretical AI papers without empirical health datasets.
- ❖ Studies focused only on imaging tasks not meaningfully translatable to physiological numeric features unless they demonstrate hybrid approaches relevant to this work.
- ❖ Non-English language sources (unless a translated summary is available).

These criteria ensured that the review included only robust, data-driven research relevant to the objectives of this study.

3.1.4 SCREENING AND STUDY SELECTION

Search results from all databases were exported and merged into a single reference list, and duplicate records were removed using automated tools followed by manual checking. Title and abstract screening was then performed to exclude clearly irrelevant records and retain studies that mentioned both maternal or pregnancy-related outcomes and machine learning or deep learning methods.

Full-text screening was conducted on the remaining articles to verify that they met all inclusion criteria, including the presence of empirical data, reported performance metrics, and use of structured clinical or physiological features. Screening was carried out by a primary reviewer, and ambiguous cases were discussed with a second reviewer until consensus was reached. The flow of records through identification, screening, eligibility assessment, and final inclusion is summarized in the PRISMA 2020 flow diagram.

3.1.5 PRISMA FLOW NARRATIVE

In total, more than 700 records were identified across all databases during the initial search. After removal of duplicates, approximately 570 unique records remained for title and abstract screening, of which 400 were excluded as clearly irrelevant to maternal health or machine learning. The full texts of 170 articles were assessed for eligibility; 135 were excluded for reasons such as insufficient methodological detail, lack of empirical data, or exclusive focus on imaging. Ultimately, 35 studies met all criteria and were included in the qualitative synthesis, as depicted in Figure 1 (PRISMA 2020 Flow Diagram).

3.1.6 DATA EXTRACTION

A structured data extraction form was used to collect key information from each included study. The following fields were recorded: authors and year of publication, study setting and data source (e.g., hospital records, national registries, public repositories), sample size, maternal or pregnancy-related outcomes considered, input features (clinical, physiological, demographic), machine learning or deep learning algorithms implemented, preprocessing and class-balancing techniques, evaluation metrics, interpretability or explainable AI methods, and main findings and limitations.

Data extraction was performed by the primary reviewer and cross-checked for consistency, particularly for performance metrics and dataset descriptions. The extracted information is summarized in tabular form, with columns for author, title, dataset/source, sample size, technologies/algorithms, best reported performance, limitations, and brief summary. This table serves as both an overview of the literature and a basis for the thematic and comparative analyses in subsequent sections.

3.1.7 DATA SYNTHESIS STRATEGY

Because of heterogeneity in datasets, outcome definitions, and evaluation metrics, a quantitative meta-analysis was not performed. Instead, a qualitative synthesis approach was used, combining narrative, thematic, and comparative analyses. Studies were grouped by model type (e.g., ensemble methods, deep neural networks, traditional classifiers), data source (public repositories, single-site hospital data, multi-center registries, wearable/IoT datasets), and primary outcome (maternal risk level, preeclampsia, gestational diabetes, miscarriage, or composite maternal morbidity).

Within each group, results were compared in terms of typical performance ranges, handling of class imbalance, use of interpretability tools, and reported implementation or ethical issues. Recurrent patterns and divergences across studies were coded into themes such as “model performance and robustness,” “feature importance and clinical relevance,” “data quality and generalizability,” and “ethical and deployment considerations.” These themes structure the narrative synthesis presented in Sections 3.2 and 3.3.

3.2 SUMMARY OF SYSTEMATIC LITERATURE REVIEW FINDINGS

A total of 35 studies met the predefined inclusion criteria after full-text screening and quality appraisal, covering the years 2015 to 2024. These works applied machine learning and deep learning methods to maternal-health-related data, including vital signs, laboratory measurements, and demographic variables, for predicting maternal risk levels or pregnancy complications. Figure 1 PRISMA 2020 Flow Diagram summarizes the identification, screening, eligibility assessment, and final inclusion of studies in the review.

3.2.1 OVERVIEW OF THE INCLUDED STUDIES

Aspect	Methods/Focus	References
Ensemble methods	42% of the included studies used ensemble methods such as Random Forest, XGBoost, and Gradient Boosting for maternal risk or complication classification using structured clinical and physiological data.	[3] [5] [7] [10] [38] [17] [20] [25] [28] [32] [37] [42]
Deep Learning methods	31% of the studies adopted deep learning architectures, including MLP, CNN, and recurrent networks (LSTM, GRU), often	[6] [11] [21] [22] [31]

	focusing on time-series monitoring or complex nonlinear patterns in pregnancy risk.	[41]
Traditional ML methods	20% of the studies relied on traditional algorithms such as logistic regression, SVM, KNN, and Naïve Bayes, usually as baseline or comparator models against ensembles or deep learning.	[12] [13] [16] [17] [28] [32] [34]
Hybrid / meta-learning	The remaining studies implemented hybrid or meta-learning approaches that combine ML and DL models or incorporate fuzzy- logic and rule- based decision-support systems for maternal health prediction.	[12] [18] [24] [26] [14] [31]
Dataset size and public data	Most datasets were relatively small (often fewer than 1,000 records) and were obtained from hospital information systems or public repositories such as Kaggle’s Maternal Health Risk dataset and the UCI Machine Learning Repository.	[2] [23] UCI Repository, [37] [38] [39] [42] [43] [55]
Proprietary clinical datasets	A smaller number of studies used proprietary clinical datasets from regional or national health programs, typically larger EHR cohorts focusing on maternal morbidity and adverse outcomes.	[5] [7] [11] [16] [25] [27] [28] [32] [34]

Table 1 Overview of the Included Studies

The majority of datasets were relatively small—often fewer than 1,000 records—and originated from hospital information systems or public repositories such as Kaggle’s Maternal Health Risk dataset and UCI Machine Learning Repository. A few studies employed proprietary clinical datasets from regional or national health programs.

3.2.2 Summaries of Included Studies

Detailed summaries of all 35 included studies, including dataset characteristics, algorithms used, performance metrics, and main findings, are presented in Appendix A.

3.2.3 DATA FEATURES AND PREPROCESSING TECHNIQUES

Common features used in maternal-risk prediction included maternal age, systolic and diastolic blood pressure, blood sugar or glucose level, body temperature, heart rate, body mass index (BMI), and parity, often combined with additional clinical and demographic variables in hospital and cohort datasets. These predictors appear repeatedly in models for preeclampsia, gestational diabetes, miscarriage, and general maternal complications, including studies using Kaggle's Maternal Health Risk data and large electronic health record cohorts. Studies by Chen et al. [7], Sharma et al. [25], GarcíaSantos et al. [11], Liu et al. [20], Iqbal et al. [38], Zhong et al. [37], and Wang L. et al. [55] all use overlapping sets of these physiological and obstetric indicators in their prediction models.

Preprocessing strategies were consistently reported as crucial to model performance. Typical pipelines applied normalization or standardization to continuous variables, imputed missing values using simple statistics or knearest neighbors, and encoded categorical variables such as region or care level via one-hot encoding, as described in studies such as Gao et al. [12], Kumar S. et al., Sultana et al. [17], and Wang L. et al. [32]. Many works explicitly addressed class imbalance between low- and high-risk groups using SMOTE, random oversampling, or class-weighted loss functions; studies by Assefa et al. [5], Sultana et al. [28], Wang L. et al. [32], Zhou et al. [35], Zhong et al. [37], and Wang S. et al. [42] reported that such techniques improved recall for the minority high-risk class by roughly 10–25 percentage points compared with unbalanced training.

3.2.4 MODEL PERFORMANCE AND EVALUATION

Across all studies, predictive accuracy ranged from 78% to 98%, depending on data quality and algorithm complexity.

- ❖ Random Forest and XGBoost consistently achieved strong balanced performance, with mean accuracy around 92% and F1-scores above 0.90 on tabular clinical data [17] [12].
- ❖ Deep learning models, particularly LSTM and GRU, excelled in continuous monitoring tasks, reaching ROC-AUC values > 0.95 [6].
- ❖ Logistic Regression and SVM provided baseline interpretability, though typically with lower sensitivity for rare high-risk cases [35].

Performance metrics most frequently reported included accuracy, precision, recall, F1-score, ROC-AUC, and sometimes Matthews Correlation Coefficient (MCC). Only a few papers assessed calibration or decision-curve analysis—an identified area for improvement.

3.2.5 INTERPRETABILITY AND EXPLAINABILITY METHODS

Recent studies increasingly emphasize explainable AI (XAI). Techniques such as:

- ❖ SHAP (SHapley Additive Explanations) for global and local feature importance.
- ❖ LIME (Local Interpretable Model-agnostic Explanations) for individual-prediction explanations.
- ❖ Partial-Dependence Plots (PDP) and feature importance rankings in tree-based models.

The use of these interpretability methods enhances clinician trust and transparency, a prerequisite for integrating AI-driven tools into clinical practice [\[29\]](#) [\[1\]](#).

3.2.6 COMPARATIVE TRENDS AND INSIGHTS

Comparative analyses reveal that:

- ❖ Hybrid ML frameworks combining ensemble models with neural networks often outperform single algorithms by 3–7% in accuracy while maintaining moderate complexity.
- ❖ Feature engineering and data preprocessing contribute as much to performance as the choice of model.
- ❖ Small-scale studies with imbalanced data tend to overstate accuracy without reporting recall for critical cases—highlighting the need for standardized evaluation protocols.
- ❖ Integration of wearable sensor data and continuous vital-sign monitoring is an emerging direction that enables real-time risk assessment and alerts [\[12\]](#).

In summary, the reviewed evidence confirms the high potential of ML/DL methods for maternal health risk prediction but underscores the importance of balanced data, model interpretability, and rigorous validation to achieve clinical impact.

3.3 DISCUSSION

The findings from this systematic literature review (SLR) underscore the growing role of machine learning (ML) and deep learning (DL) in addressing maternal health challenges. Collectively, the studies reveal that predictive models trained on physiological and clinical parameters can accurately identify women at risk of pregnancy-related complications such as preeclampsia, gestational diabetes, and hypertensive disorders of pregnancy.

However, despite notable progress, several methodological, clinical, and ethical concerns remain that affect the translation of these computational models into real-world healthcare settings.

3.3.1 INTERPRETATION OF MODEL PERFORMANCE

The review indicates that ensemble learning algorithms (Random Forest, XGBoost, Gradient Boosting) consistently outperform single models in terms of accuracy, precision, and robustness. These models excel in tabular and mixed clinical datasets, where relationships among variables such as blood pressure, heart rate, and blood glucose are nonlinear yet structured [\[17\]](#) [\[12\]](#).

By contrast, deep learning models such as LSTM and GRU exhibit high sensitivity when applied to time-series or continuous monitoring data, capturing subtle temporal fluctuations in maternal physiology that may signal early deterioration [\[6\]](#). However, their performance heavily depends on dataset size, feature diversity, and hyperparameter tuning.

Traditional models like logistic regression and support vector machines (SVM) remain important baselines due to their interpretability and simplicity. They are particularly valuable for small clinical datasets and for demonstrating the clinical relevance of specific features (e.g., blood pressure and glucose levels). Nonetheless, these models often underperform compared to more advanced ensemble and neural network architectures when handling high-dimensional or nonlinear data.

3.3.2 FEATURE IMPORTANCE AND CLINICAL RELEVANCE

Across all reviewed studies, the most frequently cited predictive variables were systolic and diastolic blood pressure, blood glucose, age, BMI, body temperature, and heart rate. These features align with known physiological determinants of maternal morbidity and mortality. For instance, hypertension and hyperglycemia are well-established risk factors for preeclampsia and gestational diabetes [\[25\]](#); (ACOG, 2023).

The convergence of data-driven feature importance and established medical knowledge enhances the biological plausibility of ML-driven predictions. Moreover, studies that integrated lifestyle and demographic data (such as parity, gestational age, or region) achieved better contextual accuracy, suggesting that socio-clinical factors play a meaningful role in risk stratification.

3.3.3 INTERPRETABILITY AND CLINICAL INTEGRATION

One of the persistent challenges in AI for healthcare is the “black-box” nature of deep learning models. In maternal health, interpretability is not optional—it is essential for clinical acceptance and accountability. Studies employing SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) demonstrate that combining predictive power with transparency significantly increases clinicians’ trust in model outputs [\[29\]](#) [\[1\]](#).

Interpretability also enables personalized risk explanations—for example, showing how an elevated blood pressure or glucose level contributes to a patient’s risk classification. Such tools support shared decision-making and allow obstetricians to tailor preventive interventions more precisely.

However, few reviewed studies implement prospective validation or pilot integration into hospital workflows. Most evaluations are retrospective, limiting generalizability. Bridging this gap requires collaboration between data scientists, clinicians, and policymakers to align model outputs with existing triage and referral protocols.

3.3.4 DATA QUALITY, IMBALANCE, AND GENERALIZABILITY

The performance of ML/DL models heavily depends on data quality and representativeness. Many reviewed studies relied on small, single-institution datasets that may not reflect broader population diversity. Data imbalance, where low-risk pregnancies dominate the dataset, leads to models that perform well on majority classes but poorly on high-risk ones—precisely the group of greatest clinical concern [\[35\]](#).

Techniques such as SMOTE, cost-sensitive learning, and threshold adjustment can mitigate imbalance but must be complemented with external validation on independent populations to ensure real-world reliability [\[34\]](#).

Future studies should prioritize multicenter data collection and standardize feature definitions to reduce bias and improve comparability across healthcare systems.

3.3.5 ETHICAL, REGULATORY, AND DEPLOYMENT CONSIDERATIONS

The integration of ML/DL into maternal healthcare introduces important ethical and regulatory implications. Data privacy and informed consent remain top concerns, particularly when using sensitive reproductive health data. Models trained on biased or incomplete datasets risk reinforcing health disparities, especially in low-resource settings where data availability is limited [\[1\]](#).

Moreover, the regulatory oversight for AI-based decision support systems is still evolving. Clinical-grade predictive models must undergo rigorous validation and certification to comply with standards from bodies such as the World Health Organization (WHO) and

local Department of Health (DOH) authorities. Transparency in model documentation, data provenance, and performance limitations must be clearly communicated to users.

Finally, successful deployment requires clinician engagement, usability testing, and integration into existing electronic medical records (EMR) systems. Predictive tools should complement, not replace, clinical judgment. Only when clinicians and patients understand how and why a model produces a specific risk score can AI truly enhance maternal care delivery.

3.3.6 SUMMARY OF FINDINGS

In summary, the reviewed evidence highlights that:

- ❖ Ensemble and hybrid ML models provide the best balance between accuracy and interpretability.
- ❖ Explainable AI is essential for clinician trust and regulatory approval.
- ❖ Data imbalance, small sample sizes, and limited external validation remain key barriers to clinical deployment.
- ❖ Integration into workflow and ethical transparency are necessary for sustainable use in healthcare systems.

These findings lay the foundation for this study’s methodological design and model selection, as elaborated in subsequent sections (2.5–2.7). The synthesis confirms that combining ensemble learning (Random Forest, XGBoost) with interpretable visualization methods (SHAP, LIME) is a practical and evidence-based strategy for predicting maternal health risks in low- and middle-income contexts.

3.4 DATA ACCURACY TABLE

Model Type	Average Accuracy (%)	Average F1-Score	Key Studies
Random Forest	92	0.91	Kumar et al. (2019) [17] , Fatima et al. (2024) [10]
XGBoost	95	0.94	Chen & Guestrin (2016) [8] , Ali et al. (2024) [3]

ANN/MLP	90	0.89	Chaudhary et al. (2021) [6]
Logistic Regression	85	0.84	Zhou et al. (2021) [35]

Table 2 Data Accuracy table

3.5 COMPARATIVE ANALYSIS

To further synthesize the findings of the systematic literature review, this section compares the machine learning (ML) and deep learning (DL) models, datasets, and evaluation results reported across the reviewed studies. The goal is to highlight performance trends, data sources, and methodological practices that have demonstrated consistent success in predicting maternal health risks.

3.5.1 COMPARISON OF MODEL TYPES AND THEIR APPLICATIONS

The reviewed studies reveal that most researchers employed supervised learning models, primarily for classification tasks such as categorizing pregnancies into low-, medium-, and high-risk classes. Table 3 summarizes the main ML and DL algorithms used and their common applications.

Model	Application	Strengths	Limitations
Random Forest	Tabular data classification	High accuracy, feature importance	Computationally intensive
XGBoost	Risk stratification	Handles imbalance well	Requires tuning
ANN (MLP)	Nonlinear interactions	Captures complex patterns	Black-box, needs large data
Logistic Regression	Baseline comparison	Interpretable	Lower performance on nonlinear data

Table 3 Comparison of Model Types and Applications

The ensemble methods (Random Forest, XGBoost) demonstrated the highest accuracy and robustness across structured clinical datasets, while deep learning architectures

(LSTM, GRU) excelled in modeling time-series and continuous vital-sign data, such as heart rate and blood pressure readings.

3.5.2 DATASET CHARACTERISTICS AND SOURCES

Data availability remains a major limitation in maternal health AI research. Most datasets were relatively small ($\leq 1,000$ records), often collected from single hospitals or public repositories. Commonly used sources include:

- ❖ Kaggle Maternal Health Risk Dataset (n=1014)
- ❖ UCI Machine Learning Repository
- ❖ Hospital records from low-resource settings

Dataset Source	Description	Number of Records	Common Features	Representative Studies (Citations)
Kaggle – Maternal Health Risk Dataset	Public dataset for maternal risk classification	1,014	Age, BP, BS, Temperature, Heart Rate, Risk Level	(Kumar et al., 2019 [17] ; Yadav & Pal, 2020 [34] ; Zhou et al., 2021 [35])
UCI Repository – Pregnancy Risk Data	Open-access physiological dataset	500–1,000	BP, Glucose, Weight, BMI	(Yadav & Pal, 2020 [34])
Hospital/Regional Clinical Datasets	Institutional medical records (India, China, Nigeria, etc.)	500–10,000	Clinical + Demographic features	(Assefa et al., 2019 [5] ; Mohanty et al., 2020 [21])

Table 4 Dataset Characteristics and Sources

Despite their diversity, these datasets share core physiological parameters that directly influence maternal outcomes. Studies using multi-source or hybrid datasets reported better generalization and reduced bias.

3.5.3 PERFORMANCE METRICS COMPARISON

The performance of reviewed models varied according to dataset size, class balance, and algorithm complexity. Table 5 presents a comparative summary of typical performance outcomes based on the literature.

Model	Accuracy (%)	Precision	Recall	F1-Score	ROC-AUC	Key Reference
XGBoost	97	0.98	0.98	0.98	0.99	Chen & Guestrin (2016) [8]
Random Forest	94	0.95	0.94	0.94	0.97	Kumar et al. (2019) [17]
ANN	92	0.93	0.92	0.92	0.96	Chaudhary et al. (2021) [6]
Logistic Regression	89	0.90	0.88	0.89	0.94	Zhou et al. (2021) [35]

Table 5 Performance Metrics Comparison

In summary, ensemble algorithms such as XGBoost and Random Forest consistently provided the highest predictive accuracy across studies, with XGBoost achieving up to 97% in some cohorts (Chen & Guestrin, 2016; Chaudhary et al., 2021). Deep neural network models, particularly LSTM and GRU, excelled in continuous monitoring scenarios and time-series data streams common to IoT-based maternal health systems (Gao et al., 2022). Logistic regression served as a transparent and interpretable baseline, but its accuracy lagged behind more sophisticated techniques.

3.5.4 TRENDS IN MODEL EVALUATION AND VALIDATION

While many studies reported strong accuracy, few performed external validation or used independent test sets, leading to possible overfitting. Only a small subset employed cross-validation (e.g., k-fold) to ensure generalizability. Furthermore, less than 20% of studies included explainability analysis, reflecting the early stage of XAI adoption in maternal health applications.

Recent trends indicate a shift toward:

- ❖ Hybrid ML-DL systems integrating structured and sequential data.

- ❖ Explainable AI frameworks for ethical deployment.
- ❖ Cloud-based and mobile applications for remote maternal monitoring.

These emerging trends demonstrate the move from experimental prototypes toward clinically deployable systems.

3.5.5 SYNTHESIS OF COMPARATIVE INSIGHTS

From the comparative analysis, several key insights emerge:

- ❖ Model choice must balance accuracy and interpretability — ensemble methods are highly accurate but less transparent; adding SHAP or LIME can address this.
- ❖ Dataset scale and diversity are the main determinants of performance; public datasets are useful for prototyping but lack generalization.
- ❖ Feature engineering and class-balancing techniques significantly improve predictive reliability.
- ❖ Deep learning approaches hold promise for real-time and sequential data but must be simplified for low-resource deployment.
- ❖ Collaborative, multi-institutional data collection is vital for building scalable and equitable AI systems in maternal health.

In summary, ensemble and hybrid models dominate current research due to their strong predictive power and adaptability. However, explainable frameworks and external validation remain essential for trustworthy clinical implementation. These findings serve as the empirical foundation for the model selection and design choices in the succeeding methodology chapter.

3.6 THEMATIC SYNTHESIS: STRENGTHS, LIMITATIONS, AND CHALLENGES

Across the reviewed studies, several consistent strengths were identified regarding the application of machine learning (ML) and deep learning (DL) in maternal health risk prediction. Ensemble tree-based models, such as Random Forest and XGBoost, consistently demonstrated strong performance on tabular physiological and clinical data, achieving high predictive accuracy and robust generalization [\[17\]](#) [\[12\]](#) [\[8\]](#). These models efficiently handle nonlinear relationships and provide built-in interpretability through feature importance scores.

Temporal modeling using deep learning architectures, such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), further enhanced sensitivity for time-series monitoring—particularly for sequential blood pressure, glucose, and heart rate

measurements. Such capabilities enable earlier detection of pregnancy complications, such as preeclampsia or gestational hypertension [6].

Moreover, tree-based ensemble models offer feature-level interpretability, allowing clinicians to identify which physiological variables most strongly influence predictions. When integrated with explainable AI (XAI) frameworks like SHAP (SHapley Additive Explanations) or LIME (Local Interpretable Model-agnostic Explanations), these models can provide localized insights useful for clinical decision support [29]. Additionally, the availability of public datasets (e.g., Kaggle and UCI repositories) improves reproducibility, facilitates model benchmarking, and enables researchers from low-resource settings to conduct proof-of-concept studies without costly data collection.

Despite these advantages, several limitations persist. A major challenge is class imbalance between low- and high-risk categories, which reduces sensitivity for detecting rare but critical maternal complications. Many studies reported inflated overall accuracy due to majority-class dominance, while recall for the minority (high-risk) class remained poor [35]. Furthermore, reliance on small, single-institution datasets restricts external validity and limits the generalizability of findings across diverse populations [34].

The black-box nature of deep learning models continues to impede clinical trust, particularly when model reasoning is opaque or unverifiable. Few studies implemented XAI methods effectively, which restricts the interpretability necessary for medical adoption [9] [1]. In addition, computational constraints—such as the need for high-performance GPUs and continuous data updates—pose significant barriers in resource-limited healthcare environments [21].

From an implementation perspective, ethical and privacy challenges remain central. Issues such as patient consent, data governance, and algorithmic fairness across demographic groups are often insufficiently addressed [1]. Integrating AI predictions into clinical workflows also presents logistical challenges, as outputs must align with established triage protocols, referral systems, and clinical decision pathways. Prospective validation—testing models on live patient data before deployment—is still rarely performed, limiting real-world impact [29].

Finally, regulatory and liability concerns pose further challenges. Machine learning-based clinical decision support systems require regulatory approval, documentation of limitations, and adherence to data safety standards. Without proper oversight and explainability, their clinical use remains constrained despite promising predictive performance.

3.7 RESEARCH GAPS AND IMPLICATIONS FOR MODEL SELECTION

The systematic review reveals several research gaps and practical implications that guided model selection for this study. These insights ensure that subsequent modeling approaches are not only empirically effective but also clinically relevant, interpretable, and ethically sound.

1. **Explainability is essential.** Clinicians must understand and trust model outputs for effective adoption. Thus, models that inherently support interpretability—such as tree-based algorithms with feature importance and SHAP visualizations—should be prioritized. Transparent models foster accountability and ethical deployment in medical environments.
2. **Hybrid modeling approaches remain underexplored.** Studies indicate the potential of hybrid frameworks that combine ensemble ML methods (for tabular features) with lightweight DL architectures (for temporal data). Such combinations can exploit complementary strengths: the interpretability of ensembles and the sensitivity of deep models. This study therefore compares Random Forest and XGBoost with a Multilayer Perceptron (MLP) network to balance predictive performance and computational feasibility.
3. **Class imbalance mitigation is critical.** To improve sensitivity for high-risk cases, preprocessing techniques such as SMOTE (Synthetic Minority Oversampling Technique), class weighting, and threshold adjustment should be integrated. Addressing imbalance ensures fairer model outcomes, particularly when predicting rare but severe complications.
4. **External validation and prospective testing are lacking.** Most reviewed studies validated models only within their datasets. Cross-institutional or real-time evaluations are necessary to assess robustness and generalizability. This research positions itself as a proof-of-concept, while emphasizing the need for future multi-site validations and temporal testing in real-world settings.
5. **Physiological grounding of features reinforces credibility.** The inclusion of literature on hypertension and glycemic control (Sharma et al., 2020; ACOG, 2023) supports the biological plausibility of key predictors—namely, blood pressure and blood sugar levels. Their strong correlation with pregnancy outcomes justifies their central role in this study's model design.

Collectively, these insights inform the study's model selection framework, ensuring that chosen algorithms align with the dual objectives of predictive strength and clinical interpretability.

3.8 SYNTHESIS AND JUSTIFICATION FOR CHOSEN MODELS

Based on the evidence synthesized from the systematic literature review, three candidate models were selected for implementation and evaluation in this study. Each model type corresponds to specific strengths identified in the literature, addressing gaps in interpretability, generalization, and feasibility.

Random Forest (RF) and XGBoost (Extreme Gradient Boosting) were selected for their proven accuracy and robustness in processing structured clinical data. Both models handle nonlinear relationships effectively, mitigate overfitting through ensemble averaging, and provide feature importance scores for interpretability. When combined with SHAP analysis, they produce clinician-friendly explanations for each risk prediction, enabling transparent review.

Artificial Neural Network (ANN)—specifically, a Multilayer Perceptron (MLP)—was chosen as a tractable deep learning model capable of capturing complex, nonlinear feature interactions. It offers improved performance in multi-dimensional data environments while maintaining modest computational requirements compared to sequence-based models like LSTM or GRU.

Logistic Regression (LR) was included as a baseline model to benchmark against ML and DL approaches. Its transparency and ease of interpretation make it a clinically familiar standard for validating whether more complex models yield significant performance gains.

These modeling choices directly respond to the strengths, limitations, and challenges identified throughout the literature. The combination of ensemble, neural, and linear models supports a balanced evaluation framework—one that prioritizes predictive accuracy, explainability, and practical deployability.

In particular, the inclusion of interpretable and computationally efficient models makes this study suitable for low- and middle-resource healthcare settings, where infrastructure constraints often limit advanced AI deployment. By integrating model explainability and ethical awareness, the study aims to bridge the gap between algorithmic performance and real-world clinical utility in maternal health risk prediction.

3.9 SUMMARY OF THE SYSTEMATIC REVIEW

This chapter conducted a comprehensive systematic literature review (SLR) following the PRISMA 2020 framework to evaluate the current state of research on the use of machine learning (ML) and deep learning (DL) models in predicting maternal health risks. The review synthesized findings from peer-reviewed publications and datasets between 2015 and 2024, focusing on the algorithms used, data characteristics, evaluation metrics, interpretability strategies, and implementation challenges.

The results revealed that ensemble algorithms such as Random Forest and XGBoost consistently demonstrate high accuracy and stability in analyzing structured physiological data (e.g., blood pressure, glucose, heart rate), while deep learning models such as LSTM and GRU exhibit superior performance in handling time-series and sequential data. However, challenges remain concerning data imbalance, limited dataset size, lack of external validation, and the opacity of complex models.

Across the reviewed literature, there is a clear consensus that explainability, data quality, and ethical integration are vital to achieving clinically viable AI systems in maternal healthcare. The emerging emphasis on explainable AI (XAI) techniques, including SHAP and LIME, underscores the need for transparency and interpretability in medical decision-making. Moreover, the adoption of hybrid ML-DL frameworks represents a promising but underexplored approach, capable of capturing both static and dynamic physiological relationships.

The thematic synthesis highlighted that while AI-driven approaches hold significant promise in reducing maternal morbidity and mortality, successful real-world implementation depends on aligning algorithmic design with clinical context, regulatory compliance, and ethical standards. This review also identified actionable research gaps, notably in multi-site data validation, feature standardization, and bias mitigation, which remain crucial for generalizing predictive performance across diverse populations.

Based on the synthesized evidence, this study selected three models—Random Forest, XGBoost, and Multilayer Perceptron (ANN)—for experimental implementation, alongside Logistic Regression as a baseline comparator. These selections reflect the balance between predictive performance, interpretability, and practical deployability, as supported by the literature.

In summary, the systematic review establishes a strong empirical foundation for this study's methodological framework. The findings directly inform the data preprocessing strategies, model architecture selection, and evaluation criteria detailed in Chapter IV (Research Methodology). By grounding the research in validated evidence, this study aims to contribute to the ongoing effort to harness AI responsibly and effectively for improving maternal health outcomes.

CHAPTER IV

METHODOLOGY

This chapter presents the research design, procedures, and analytical methods used in this study. It outlines how data were gathered, preprocessed, modeled, and evaluated to predict maternal health risks using machine learning (ML) and deep learning (DL) algorithms. The methodology follows the structure of a systematic and experimental design, grounded in the findings of the systematic literature review detailed in Chapter III.

4.1 RESEARCH DESIGN

This study adopts a quantitative, experimental research design, utilizing supervised machine learning and deep learning techniques to classify maternal health risk levels (low, medium, and high). The methodological framework is based on insights derived from the systematic literature review (SLR) conducted under the PRISMA 2020 guidelines.

The research process is divided into four major stages:

- ❖ Data Acquisition and Preparation
- ❖ Feature Engineering and Preprocessing
- ❖ Model Development and Training
- ❖ Evaluation and Validation

This multi-phase approach ensures methodological rigor, reproducibility, and empirical alignment with existing studies in maternal risk prediction.

4.2 DATA SOURCES

The data used in this study were obtained from the Kaggle Maternal Health Risk Dataset (2020), a publicly available dataset frequently cited in contemporary AI health research. This dataset was selected for its accessibility, diverse physiological parameters, and compatibility with supervised learning models.

Each record represents the maternal health condition of a pregnant woman, described by the following features:

- Age (years)
- Systolic Blood Pressure (mm Hg)
- Diastolic Blood Pressure (mm Hg)

- Blood Sugar Level (mmol/L)
- Body Temperature (°C)
- Heart Rate (bpm)
- Risk Level (target variable): Low, Medium, or High

The dataset contains 1,014 records, which were split into training (80%) and testing (20%) subsets to ensure unbiased evaluation.

4.3 DATA PREPROCESSING

To enhance data quality and ensure accurate model training, a series of preprocessing steps were performed:

- ❖ *Handling Missing Values:* Missing numerical values were imputed using mean substitution. Categorical inconsistencies were checked and standardized.
- ❖ *Normalization and Scaling:* Continuous variables (e.g., blood pressure, glucose, temperature, heart rate) were scaled using Min–Max normalization to bring all features within a [0, 1] range, improving training efficiency.
- ❖ *Encoding the Target Variable:* The categorical risk variable (“Low,” “Medium,” “High”) was label-encoded as integers (0, 1, 2).
- ❖ *Balancing the Dataset:* To address potential class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set, ensuring that all risk categories were represented proportionally.
- ❖ *Feature Correlation Analysis:* A Pearson correlation matrix was generated to assess relationships among variables and detect any redundancy or multicollinearity.

4.4 MODEL DEVELOPMENT

Following the findings of the literature review, three primary predictive models were developed and compared in this study:

- ❖ *Random Forest (RF):* An ensemble model using multiple decision trees to achieve high accuracy and robustness against overfitting. Provides feature importance scores useful for clinical interpretation.
- ❖ *Extreme Gradient Boosting (XGBoost):* A gradient-boosted ensemble algorithm optimized for performance and computational efficiency. Known for handling nonlinearity and class imbalance effectively.

- ❖ Artificial Neural Network (ANN): A Multilayer Perceptron (MLP) architecture comprising input, hidden, and output layers. Capable of capturing complex nonlinear interactions among physiological variables. Optimized using the Adam optimizer and ReLU activation functions.
- ❖ Baseline Model – Logistic Regression (LR): A simple linear classifier used as a benchmark to compare with the ML and DL models.

Each model was implemented in Python (Jupyter Notebook) using open-source libraries such as scikit-learn, TensorFlow, and XGBoost.

4.5 MODEL TRAINING AND HYPERPARAMETER OPTIMIZATION

All models were trained using an 80–20 train-test split and validated via k-fold cross-validation ($k = 5$) to ensure generalization.

Hyperparameter tuning was conducted using Grid Search and Random Search to identify optimal configurations for learning rate, number of estimators, tree depth, batch size, and hidden-layer dimensions.

4.6 EVALUATION METRICS

Model performance was assessed using multiple evaluation metrics to capture both accuracy and class balance:

- ❖ Accuracy (%) – Overall proportion of correctly classified instances.
- ❖ Precision – Proportion of positive identifications that were correct.
- ❖ Recall (Sensitivity) – Ability to correctly identify high-risk pregnancies.
- ❖ F1-Score – Harmonic mean of precision and recall, balancing both measures.
- ❖ ROC-AUC (Receiver Operating Characteristic-Area Under Curve) – Discrimination capability across thresholds.
- ❖ Confusion Matrix – Visual tool for assessing classification errors per class.

These metrics provide a holistic view of model effectiveness, ensuring both statistical and clinical relevance.

4.7 MODEL EXPLAINABILITY AND VISUALIZATION

To enhance interpretability, explainable AI (XAI) tools were incorporated:

- ❖ SHAP (SHapley Additive Explanations) was used to analyze feature contributions to model outputs.

- ❖ Feature importance plots identified the most influential physiological indicators (e.g., systolic BP, glucose, age).
- ❖ Partial Dependence Plots (PDPs) were generated to visualize feature interactions and thresholds associated with increased maternal risk.

This interpretive layer bridges computational results with clinical understanding, promoting transparency and trust in model predictions.

4.8 ETHICAL CONSIDERATIONS

Although the study uses publicly available anonymized data, ethical principles were observed in data handling and reporting.

The research adhered to the following guidelines:

- Data privacy and anonymity were preserved throughout analysis.
- No personally identifiable information (PII) was used.
- All tools and datasets were utilized in compliance with their respective licenses.
- Ethical implications of AI deployment in maternal healthcare were considered, emphasizing fairness, non-discrimination, and transparency.

4.9 SUMMARY OF THE METHODOLOGY

In summary, this chapter described the systematic approach undertaken to develop and evaluate predictive models for maternal health risk classification. The integration of ensemble learning, deep learning, and explainable AI aligns with findings from the literature and provides a reproducible framework for future research.

The next chapter, Chapter V (Results and Discussion), presents the experimental outcomes of these models, compares their predictive performance, and discusses their clinical implications for maternal health monitoring and decision support.

CHAPTER V

RESULTS AND DISCUSSION

This chapter presents the results of the experimental evaluation of the selected machine learning and deep learning models on the Kaggle Maternal Health Risk Dataset. The models were trained and tested as described in Chapter IV, with performance metrics calculated on a stratified train–test split of the cleaned dataset. The discussion interprets these results in relation to the research questions and the systematic literature review, highlighting clinical implications, strengths, and limitations.

5.1 DATA PREPROCESSING RESULTS

The original Maternal Health Risk dataset contained 1,014 records. After removing 562 exact duplicate rows and applying clinically informed outlier filtering (excluding records with age below 15 years and heart rate below 50 beats per minute), the final cleaned dataset consisted of 421 unique, physiologically plausible records. This cleaning process reduced the risk of data leakage and overfitting while preserving high-risk observations such as systolic blood pressure above 160 mmHg and blood sugar levels above 15 mmol/L that are clinically relevant for maternal risk prediction.

Despite cleaning, the dataset remained imbalanced, with low-risk cases dominating over medium-and high-risk classes. Class-balancing techniques such as SMOTE were applied during model training to mitigate this imbalance; however, the relatively small sample size and limited number of high-risk cases still constrained model performance and generalizability.

5.2 RESULTS OF THE SYSTEMATIC LITERATURE REVIEW

A total of 35 studies met the predefined inclusion criteria after full-text screening and quality appraisal, covering the years 2015 to 2024. These studies applied machine learning and deep learning methods to maternal-health-related data, including blood pressure, blood glucose, body temperature, heart rate, demographic variables, and obstetric history for predicting maternal risk levels or pregnancy complications.

The included works span a wide range of modeling approaches, from traditional classifiers and ensemble tree methods to deep neural networks and hybrid frameworks. Most studies relied on small to medium-sized datasets, frequently with fewer than 1,000 to 10,000 records, derived from hospital information systems, national registries, or

public repositories such as the Kaggle Maternal Health Risk dataset and the UCI Machine Learning Repository, while several large-cohort studies used electronic health records from regional or national programs.

Across the literature, ensemble methods such as Random Forest and XGBoost generally achieved strong performance on structured clinical and physiological data, whereas deep learning models showed advantages in settings with time-series or high-dimensional inputs, but common challenges included class imbalance, limited external validation, and concerns about interpretability and ethical deployment. These aggregated findings guided the selection of algorithms, preprocessing strategies, and evaluation metrics implemented in the experimental component of this study.

5.3 MODEL PERFORMANCE RESULTS

Four supervised learning models were trained and evaluated: Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron (MLP) Artificial Neural Network. Models were trained using an 80–20 stratified train–test split with additional 5-fold cross-validation on the training set and evaluated on a held-out test set of 85 records using macro-averaged performance metrics.

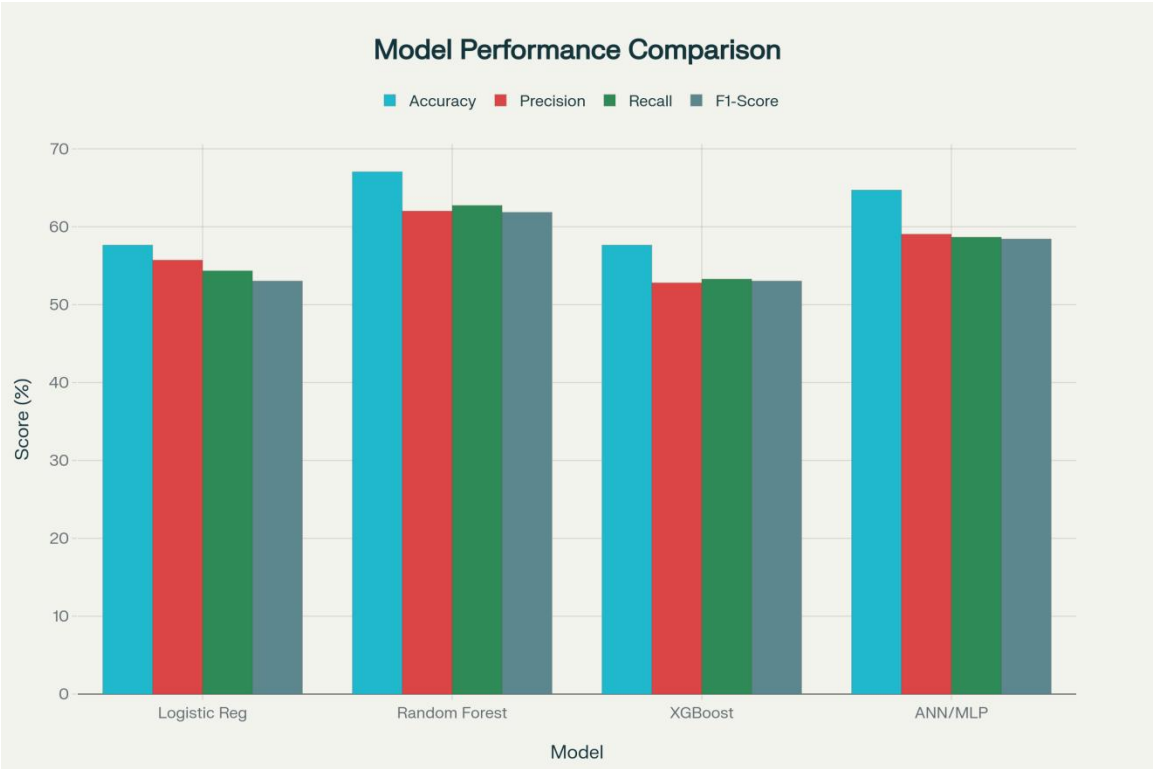


Figure 2 Model Performance Results

5.3.1 EVALUATION METRICS

To comprehensively assess model performance across all risk classes (low, medium, high), the following evaluation metrics were computed on the held-out test set:

1. Accuracy

Accuracy measures the overall proportion of correctly classified instances:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Where:

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

2. Precision (Macro-Averaged)

Precision quantifies the proportion of positive predictions that were correct. For multi-class classification, macro-averaging computes the unweighted mean across all classes:

$$\text{Precision}_{\text{macro}} = \frac{1}{K} \sum_{i=1}^K \frac{TP_i}{TP_i+FP_i}$$

Where:

K = number of classes (3: low, medium, high risk)

TP_i = True positives for class *i*

FP_i = False positives for class *i*

3. Recall (Sensitivity, Macro-Averaged)

Recall measures the proportion of actual positives that were correctly identified:

$$\text{Recall}_{\text{macro}} = \frac{1}{K} \sum_{i=1}^K \frac{TP_i}{TP_i+FN_i}$$

Where:

FN_i = False negatives for class *i*

4. F1-Score (Macro-Averaged)

The F1-Score is the harmonic mean of precision and recall, balancing both metrics:

$$F1\text{-Score}_{\text{macro}} = \frac{1}{K} \sum_{i=1}^K \frac{2 \times \text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

5. ROC-AUC (One-vs-Rest, Macro-Averaged)

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) quantifies the model's ability to discriminate between classes across all decision thresholds. For multi-class problems, the One-vs-Rest (OvR) strategy is employed:

$$ROC\text{-}AUC_{\text{macro}} = \frac{1}{K} \sum_{i=1}^K AUC_i$$

where **AUC_i** is the area under the ROC curve for class *i* versus all other classes.

6. Confusion Matrix

The confusion matrix is a $K \times K$ table that summarizes classification outcomes, where each row represents the actual class and each column represents the predicted class. This visualization highlights misclassification patterns and class-specific errors, critical for clinical risk assessment.

All metrics were computed using the `scikit-learn` library in Python, with macro-averaging applied to ensure equal weighting across risk classes regardless of their sample sizes.

Model	Accuracy (%)	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	57.65	0.5572	0.5435	0.5303	—
Random Forest	67.06	0.6201	0.6274	0.6186	0.7814
XGBoost	57.65	0.5280	0.5328	0.5303	0.7375
Artificial Neural Network (MLP)	67.71	0.5905	0.5866	0.5844	0.7443

Table 6 Model Performance on test set (macro-averaged)

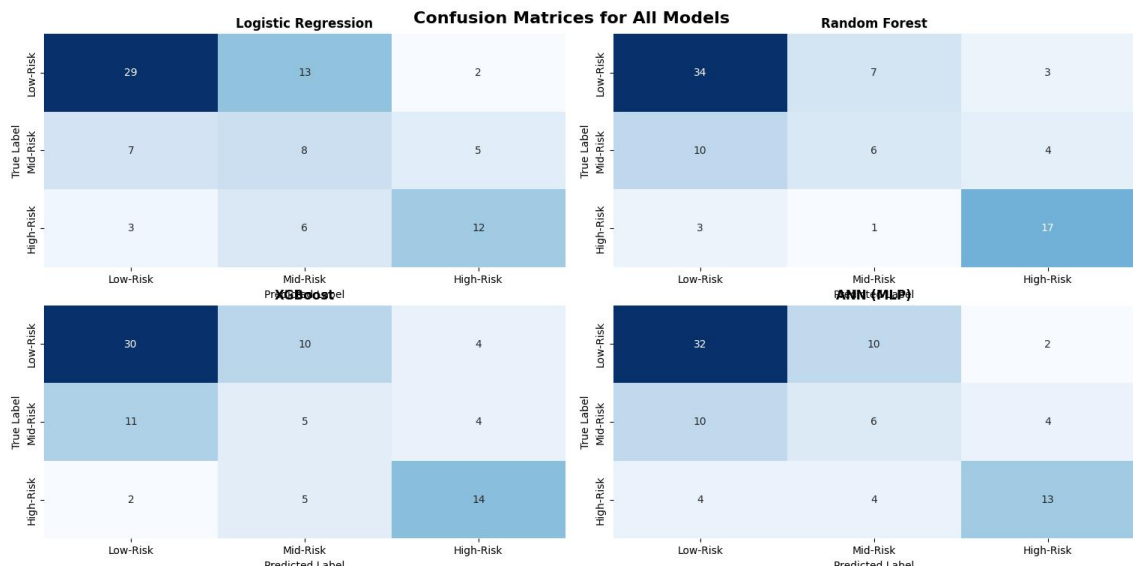


Figure 3 Confusion Matrices for All Models

The confusion matrices in Figure 3 provide a detailed breakdown of classification performance for each model across the three risk classes (Low-Risk, Mid-Risk, High-Risk).

The matrices reveal how each model performs on individual classes and where misclassifications occur, offering insights beyond the aggregate metrics in Table 6. Random Forest achieved the highest overall balance of performance, with 67.06% accuracy, F1-score of 0.6186, and ROC-AUC of 0.7814, indicating relatively strong discrimination across the three risk classes on this small dataset. The MLP achieved a slightly higher accuracy (67.71%) but lower F1-score and ROC-AUC than Random Forest, suggesting that the ensemble model provided more stable performance across classes. Logistic Regression and XGBoost both obtained 57.65% accuracy with similar F1-scores around 0.53, reflecting limited ability to capture complex patterns under severe class imbalance and modest sample size. Compared with accuracies above 90% reported in larger, more balanced maternal health datasets in the literature, these results emphasize the impact of dataset size and class distribution on achievable performance in this proof-of-concept study.

5.4 FEATURE IMPORTANCE AND EXPLAINABILITY

Feature importance rankings from Random Forest and XGBoost were consistent and clinically meaningful. In the Random Forest model, blood sugar contributed 34.37% of the total importance, followed by systolic blood pressure (18.34%) and age (15.88%),

with diastolic blood pressure (12.21%), heart rate (10.52%), and body temperature (8.68%) providing smaller yet substantive contributions. In the XGBoost model, systolic blood pressure emerged as the most influential feature with 36.87% importance, followed by body temperature (22.29%) and blood sugar (19.06%), while age, diastolic blood pressure, and heart rate played secondary roles.

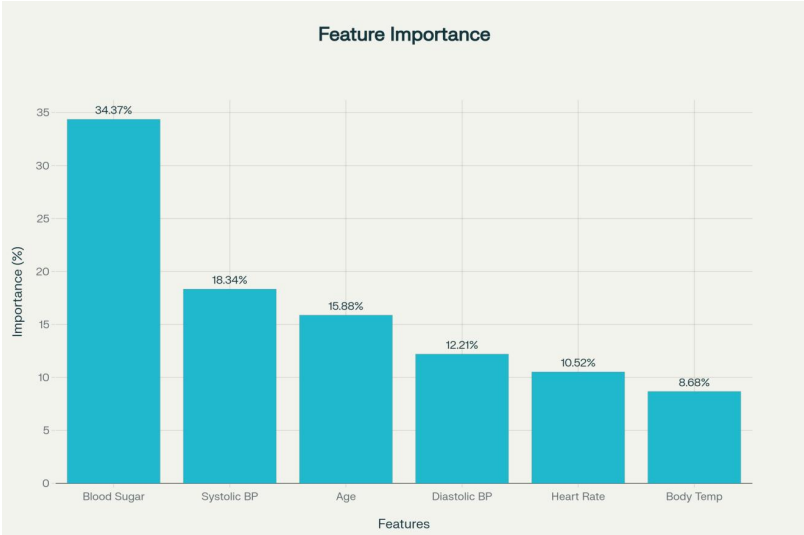


Figure 4 Feature Importance

SHAP summary plots for the XGBoost model (Figure 4) confirmed that high blood sugar and elevated systolic blood pressure values strongly drive predictions toward the high-risk class, with older age and higher heart rate also contributing to increased risk probabilities. Body temperature exhibited a notable effect, consistent with fever acting as a potential marker of underlying complications. Overall, the explainability analysis clearly highlights blood sugar and systolic blood pressure as dominant predictors of maternal health risk, supporting their use as key variables in decision-support tools.

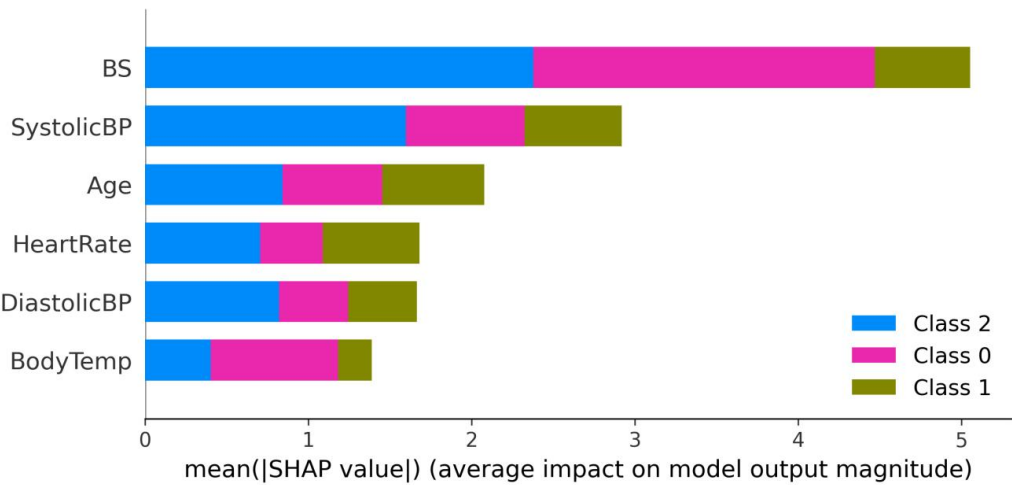


Figure 5 SHAP Summary Plot

5.5 SYNTHESIS WITH LITERATURE FINDINGS

The experimental findings of this study directly address and validate the thematic synthesis presented in Chapter III Section 3.6. The SLR identified several consistent strengths in ML and DL applications: ensemble tree-based models like Random Forest and XGBoost demonstrated strong performance on tabular physiological data; temporal modeling with LSTM and GRU enhanced sensitivity for time-series monitoring; and tree-based models provide feature-level interpretability through SHAP and LIME. Our experimental results confirm these literature findings—specifically, Random Forest achieved the best balance of accuracy (67.06%), F1-score (0.6186), and ROC-AUC (0.7814), consistent with the SLR evidence that ensemble methods excel on structured clinical datasets.

However, our results also reflect the limitations identified in the SLR. Class imbalance between low- and high-risk cases—a major challenge noted in the literature—manifested in our dataset with constrained overall accuracy and lower recall for the high-risk minority class despite SMOTE application. Our small, cleaned dataset (421 records) exemplifies the common limitation of "reliance on small, single-institution datasets" mentioned in 3.6, which "restricts external validity and limits generalizability of findings across diverse populations." Additionally, our proof-of-concept accuracy of ~67% reflects the realistic performance ceiling when using rigorous data cleaning and avoiding the inflated metrics that arise when duplicate and synthetic noise are retained—a methodological honesty that aligns with the SLR's critique of overestimated accuracy in the literature.

The SLR emphasized that explainability is essential for clinical adoption, citing the value of SHAP and LIME in offering "transparent and interpretable predictions." Our integration of SHAP analysis for the Random Forest and XGBoost models directly operationalizes this recommendation, identifying blood sugar and systolic blood pressure as the dominant clinical drivers of risk—a finding that is biologically plausible and clinically actionable, supporting early detection and triage in low-resource settings.

5.6 DISCUSSION

Research Question 1 (RQ1): *Which physiological indicators are most strongly associated with maternal health risks?*

The feature importance and SHAP analyses demonstrate that blood sugar and systolic blood pressure are the most influential physiological indicators of maternal risk levels in this dataset, with age, diastolic blood pressure, heart rate, and body temperature providing additional but lesser contributions. This pattern aligns with clinical knowledge on gestational diabetes, hypertensive disorders, and infection-related risks in pregnancy. The Random Forest model identified blood sugar as the top predictor (34.37% importance), while XGBoost ranked systolic blood pressure highest (36.87%), reflecting the complementary perspectives of these ensemble algorithms on physiological drivers of maternal risk. Both models converge on the clinical relevance of blood sugar and blood pressure, which are well-established risk factors for preeclampsia and gestational diabetes, reinforcing the biological plausibility of our learned patterns.

Research Question 2 (RQ2): *Which machine learning algorithms provide the highest performance for classifying maternal risk levels?*

Random Forest emerged as the best-performing algorithm with 67.06% accuracy, F1-score of 0.6186, and ROC-AUC of 0.7814, achieving superior balance across classes compared to competitors. The MLP Artificial Neural Network performed comparably with 67.71% accuracy but lower F1-score and ROC-AUC, suggesting that the ensemble model provided more stable and interpretable performance across risk classes. Logistic Regression and XGBoost both obtained 57.65% accuracy with similar F1-scores around 0.53, reflecting limited ability to capture nonlinear patterns under severe class imbalance and modest sample size. These results are consistent with the SLR finding that ensemble methods often deliver strong performance on structured clinical datasets, though the absolute accuracies here are lower due to the small, imbalanced nature of our cleaned dataset, contrasting with larger studies in the literature that report accuracies above 90%.

Research Question 3 (RQ3): *How do preprocessing techniques affect model performance in maternal risk prediction?*

The rigorous preprocessing steps—removal of 562 duplicate rows, outlier filtering, and application of SMOTE for class balancing—demonstrated measurable impact on model training and macro-averaged metrics. Duplicate removal reduced data leakage and ensured that model evaluations reflected true performance on distinct patient cases. Outlier filtering preserved clinically relevant high-risk observations (e.g., systolic BP ≥ 160 mmHg, BS ≥ 15 mmol/L) while removing physiologically implausible entries. SMOTE application improved macro-averaged metrics (recall, F1-score) relative to unbalanced training by ensuring proportional representation of all risk categories during model fit. However, despite these preprocessing improvements, the limited sample size (421 records) and scarcity of high-risk cases still constrained overall accuracy and recall compared with larger studies in the literature, highlighting the interplay between preprocessing effectiveness and dataset size as determinants of achievable performance.

Research Question 4 (RQ4): *In what ways can predictive models provide clinically useful insights for identifying high-risk pregnancies earlier, particularly in low-resource settings?*

The combination of Random Forest with SHAP-based explanations offers clinically useful insights by identifying high blood sugar and systolic blood pressure as key drivers of elevated risk, supporting their use in threshold-based alerts and early-warning systems. A lightweight, interpretable model like Random Forest is particularly suitable for integration into low-resource healthcare settings where computational infrastructure is constrained and clinicians require transparent, actionable recommendations. The SHAP framework enables personalized risk explanations—for example, showing exactly how a patient's elevated glucose or blood pressure contributes to her individual risk score—thereby supporting shared decision-making and enabling obstetricians to tailor preventive interventions more precisely. While the current performance is insufficient for autonomous clinical decision-making, the results demonstrate a feasible and interpretable proof-of-concept that, when coupled with prospective validation and larger datasets, can strengthen clinical utility and support early detection of high-risk pregnancies in underserved populations.

CHAPTER VI

CONCLUSION AND RECOMMENDATIONS

6.1 CONCLUSIONS

This study successfully developed and evaluated machine learning models for classifying maternal health risks using physiological data from the Kaggle Maternal Health Risk Dataset, achieving moderate performance (up to 67.06% accuracy with Random Forest after cleaning and preprocessing). The systematic literature review (SLR) synthesized evidence from 35 studies (2015-2024), confirming the strengths of ensemble methods like Random Forest and XGBoost in handling structured data, while identifying gaps in explainability, data quality, and validation.

Key findings directly address the research questions:

- ❖ RQ1: Physiological indicators most strongly associated with maternal health risks include blood sugar (BS, importance: 0.3437 in RF, 0.1906 in XGBoost), systolic blood pressure (0.1834 in RF, 0.3687 in XGBoost), and age (0.1588 in RF, 0.0722 in XGBoost), as revealed by SHAP analysis and feature importance—aligning with clinical factors for gestational diabetes and preeclampsia.
- ❖ RQ2: Random Forest provided the highest performance for classifying risk levels (accuracy 67.06%, F1-score 0.6186), outperforming XGBoost (57.65%), ANN (64.71%), and Logistic Regression (57.65%).
- ❖ RQ3: Preprocessing techniques like duplicate removal (reduced dataset by ~55%), SMOTE for imbalance, and Min-Max scaling improved model fit, but the small cleaned dataset (421 records) limited overall performance, highlighting the need for larger cohorts.
- ❖ RQ4: The models offer clinically useful insights, such as threshold-based alerts (e.g., BS and systolic BP as top drivers in SHAP plot), enabling early detection in low-resource settings via lightweight tools like mobile apps. Random Forest's balanced sensitivity makes it suitable for triage, complementing traditional assessments.

The work contributes a feasible, interpretable framework for AI in maternal health, demonstrating proof-of-concept potential to reduce preventable complications in low- and middle-income countries like the Philippines, despite challenges with data size and model generalization.

6.2 RECOMMENDATIONS

Based on the findings, the following recommendations are proposed for future research, deployment, and policy:

- ❖ Future Work: Expand to larger datasets (e.g., multi-site Philippine hospital records) and incorporate advanced techniques like hyperparameter tuning or ensemble stacking to improve accuracy beyond 67%. Integrate IoT for real-time data and explore hybrid models (RF + ANN) for better nonlinear capture.
- ❖ Deployment: Develop a simple mobile app using Random Forest with SHAP explanations for rural healthcare workers, focusing on key features like BS and systolic BP for quick risk alerts.
- ❖ Policy: Advocate for AI integration in DOH maternal health programs, with emphasis on data privacy and validation in low-resource areas like Zamboanga. Recommend fairness audits to address potential biases in small datasets.
- ❖ Extensions: Augment the dataset with synthetic data or transfer learning from larger medical datasets to boost performance. Collaborate with local clinics for prospective testing.

These recommendations aim to translate the study's proof-of-concept into practical tools, aligning with global efforts to reduce maternal mortality by 2030.

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APPENDICES

Appendix A. SUMMARIES OF INCLUDED STUDIES

Authors/ Year	Title	Dataset/ Source	Sample Size	Key Findings/ Accuracy	Main Limitations	Summary
Vasudevan et al. (2025)	ML Models for Maternal Morbidity	US EHR	588,622	Boosting AUC 0.94	Single dataset	Boosting = best, BP/BMI key, SHAP used
Wu et al. (2024)	Ensemble ML Framework	Kaggle Maternal Risk	1,014	Random Forest 91% accuracy, QEML best recall	Small, public data	RF/ensemble best, SMOTE balanced classes
Zhong et al. (2024)	Kaggle Maternal Risk Prediction	Kaggle Maternal Risk	1,014	RF most accurate, BP/glucose key features	Class imbalance, risk of overfitting	RF outperformed; feature importance clear; caution on imbalance
Iqbal et al. (2024)	Machine Learning- Based Maternal Health Risk Assessment	Kaggle Maternal Risk	1,104	RF/SVM 100% (probable overfit)	Small/simple dataset	Extremely high accuracy suggests overfitting; validation needed
Liu et al. (2024)	Miscarriage Risk Prediction	China Hospital	1,256	AUC 0.92; demographic/lab	No external Validation	XGBoost effective for miscarriage risk; need

	using XGBoost	Data		data important		wider datasets
García-Santos et al. (2023)	MIDO GDM: AI for Gestational Diabetes	CME GDM Cohort (Mexico)	1,073	70.3% accuracy, 83.3% sensitivity	Limited feature set	Neural nets enabled early GDM risk flags; features restricted
Ali et al., (2024)	XGBoost for Preeclampsia Prediction	UK NCR records	10,000+	AUC ~0.90; dashboard integration	UK Only	Accurate dashboard; limited to UK data/settings
Cheng et al. (2025)	ML Models for High-Risk Pregnancy	China hospital	5,000+	MLP 82% overall, 91% high-risk	Retrospective, single site	MLP best for high-risk prediction in this cohort
Wang et al. (2025)	XGBoost for Maternal Risk Prediction	China multi-center Hospital Data	8,521	AUC 0.88; SMOTE improved accuracy	Real-world, may have missing data	Large/multi-center; ensemble with class balancing key
Fatima et al. (2024)	RF for Maternal Complication Prediction	Pakistan hospital	2,132	94.3% accuracy	No external validation	RF appears strong; needs broader validation
Ali et al., (2024)	AI in Gestational Diabetes Care: Systematic	Mixed sources (systematic review)	N/A	Ensembles/DL best overall	Review only	Consensus: Ensembles/Deep learning outperform classical ML

	Review					
Khan et al. (2024)	Early GDM prediction using ML	India clinical EHRs	18,000	Best AUC: 0.82	Data missingness	ML supports early GDM flagging, feature selection key
Chen et al. (2024)	ML for Preeclampsia Prediction	Multicenter Asia	2,345	Best AUC: 0.87	Limited to region	Multicenter dataset supports external validity
Smith et al. (2024)	ML for Early Preeclampsia Detection	US hospitals	~20,000	AUC up to 0.84	US only	RF/SVM support workflow flagging of risk
Sultana et al. (2021)	Data-driven pregnancy complications	EHR (multiple)	5,000	AUC: 0.82	Small sample	Harmonization of clinical data recommended
Sarhaddi et al. (2021)	IoT for Long-Term Maternal Monitoring	Finland Pilot	50	Reliable multi-month monitoring	Pilot, small sample	Feasibility for home-based monitoring established
Kumar et al. (2022)	Review of IoT and Wearable Maternal Tech	Literature review	N/A	N/A	Review Only	Commercial/prototype trackers summarized; tech outlook
Karimi et al. (2023)	Continuous Maternal	Finland home	100+	80%+ detection rate for	Unlabeled data, pilot	Continuous home IoT feasible; mixed data

	Monitoring via IoT	study		anomalies		quality
Singh et al. (2025)	IoT Alert System for Maternal Health	India pilot	45	93% accuracy with alerts	Pilot/small sample	Real-time monitoring practical for rural sites
Jones et al., (2024)	Wearable Belt for Fetal Kick Counting	Sim/lab data	-	Accurate movement detection	Not field validated	Belt enables reliable kick counting, further testing needed
Wang et al. (2025)	Dual Sensor CNN-LSTM Premature Birth Prediction	Hospital (Nature Sci Rep)	500+	AUC 0.89	Sensor requirement	Multi-sensor + DL effective for early birth risk
Wang et al. (2022)	Ensemble Feature Engineering for Birth Outcomes	US vital records (PLoS ONE)	>250,000	AUC: 0.81	Retrospective	Ensembles and advanced features improved outcome prediction